

Aktiva filter, Oscillatorn och Komparatorn

Viktor Öwall



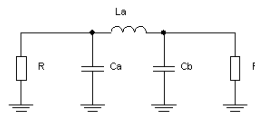
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Aktiva Filter

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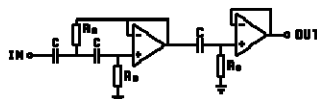
Passiva Filter

- Innehåller R, C och L
- Vanligt inom Radio
- Vid låga frekvenser blir komponenter
 - stora
 - med stora förluster
 - dyra



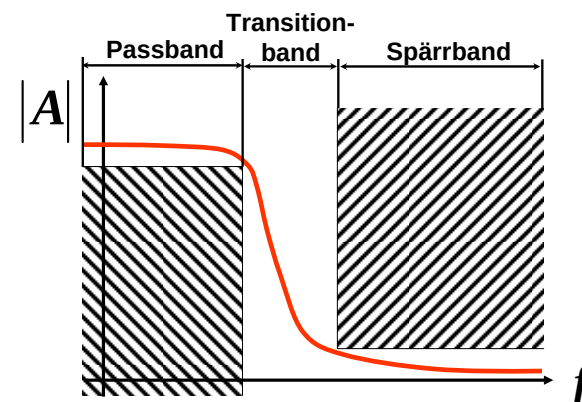
Aktiva Filter

- Innehåller R, C och aktiva element
- Vanligt vid "låga" frekvenser
- Lättare att styra
- Kan ge förstärkning.
- Kräver spänningsmatning
- Återkoppling → risk för instabilitet



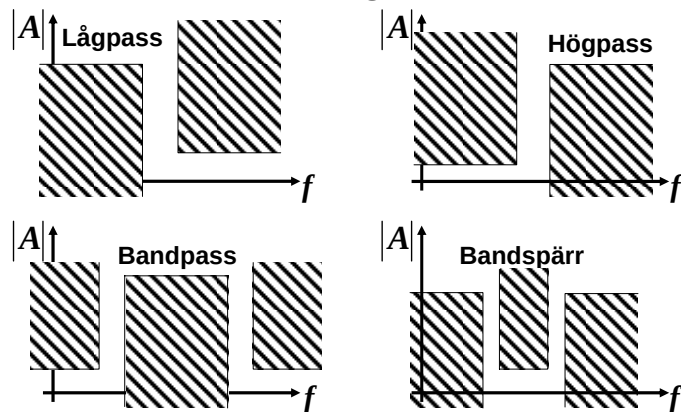
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Klassificering av filter



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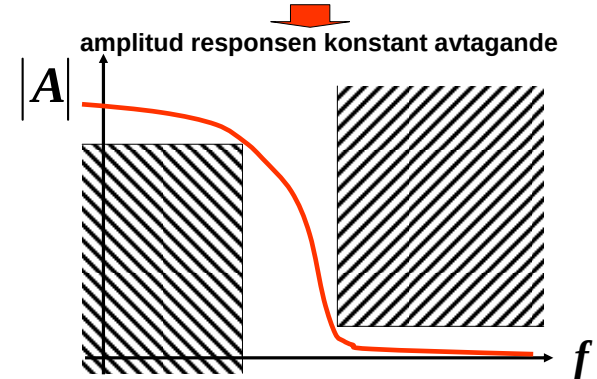
Klassificering av filter



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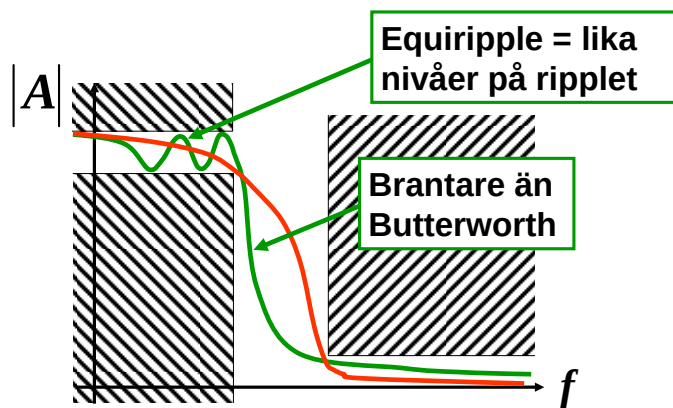
Butterworth-filter

har "maximally flat amplitude response"



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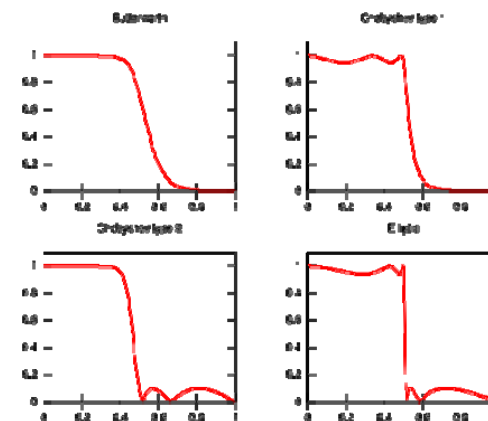
Chebyshev - filter



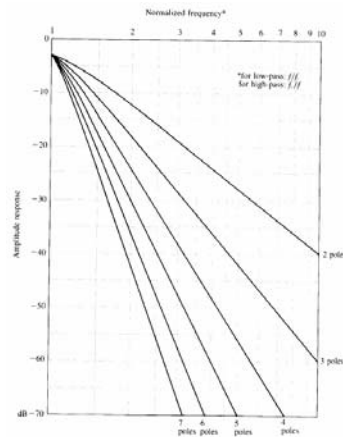
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Different Filter

http://en.wikipedia.org/wiki/Category:Filter_theory



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Bode-diagram
för
lågpassfilter
med olika
antal poler

Många poler

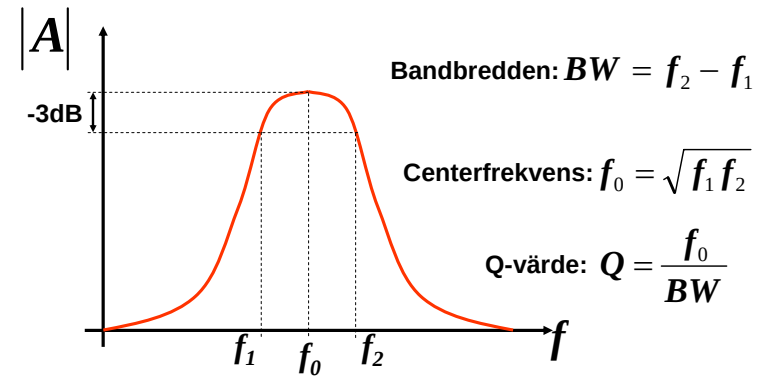


Brant filter

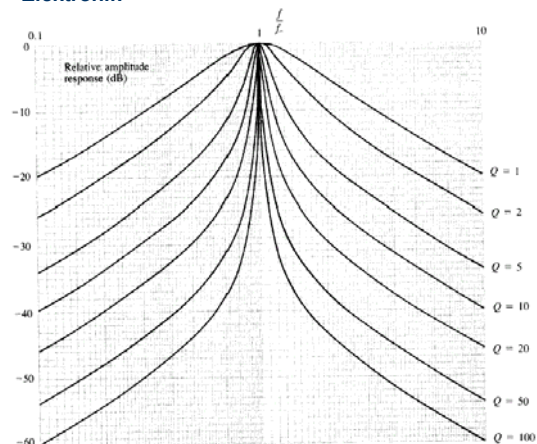
FIGURE 8-7
Sharp-band amplitude response of Butterworth filters.

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Q-värde = kvalitetsmått



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Bode-diagram
för
bandpassfilter
med olika
Q-värden

Högt Q-värde



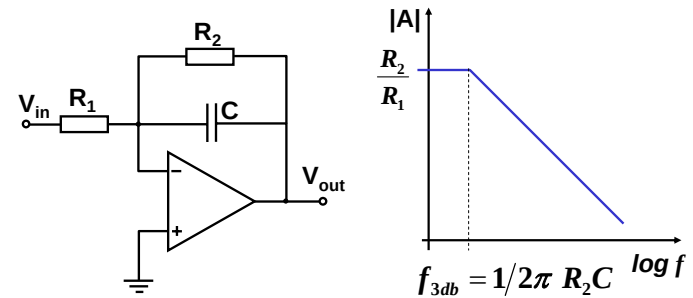
Brant filter

FIGURE 8-20
Two-pole band-pass amplitude response curves.

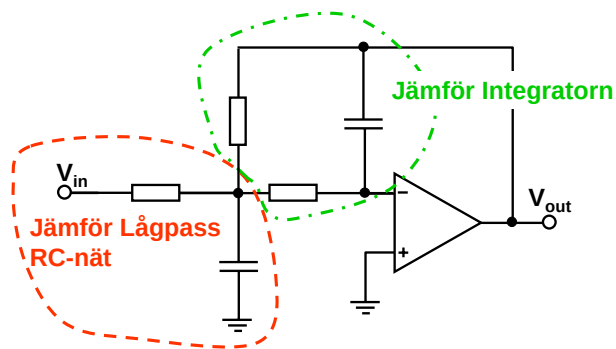
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Första ordningens lågpas

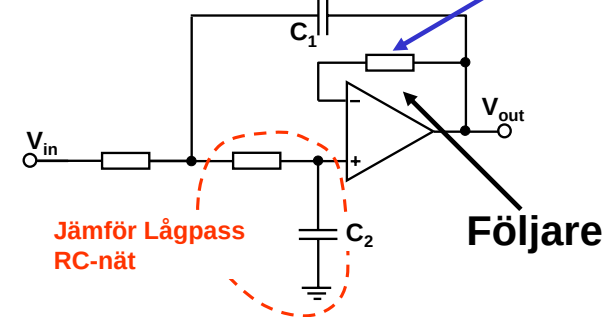
Integratorn



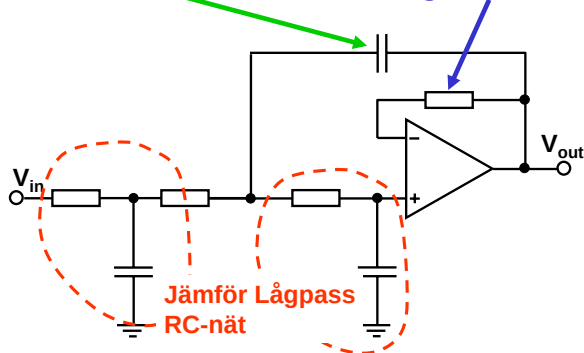
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Andra ordningens lågpasNegativ återkoppling

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Andra ordningens lågpasPositiv återkopplingNegativ återkoppling

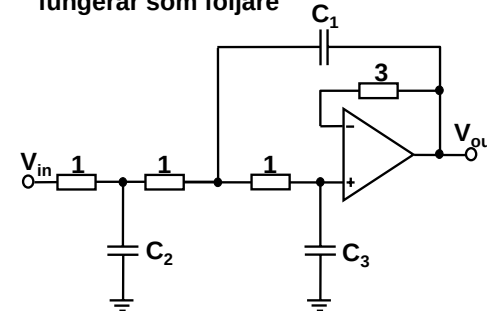
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Tredje ordningens lågpasPositiv återkopplingNegativ återkoppling

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Tredje ordningens LP – Unity Gain

Unity Gain = Förstärkarna fungerar som följare



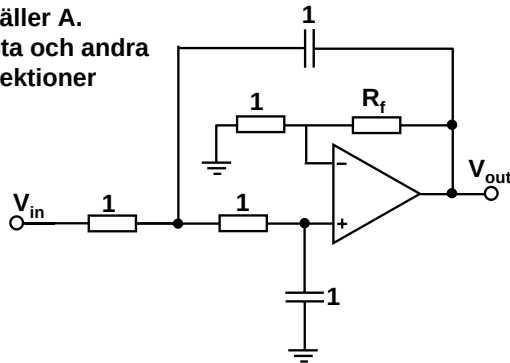
Poles	C ₁	C ₂	C ₃
2	1.414	0.7071	
3	3.546	1.392	0.2024
4	1.082	0.9241	
5	1.753	1.354	0.4214
6	1.035	0.9660	
7	1.531	1.336	0.4885

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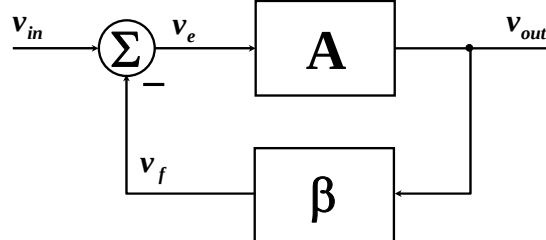
Tredje ordningens LP – Equal Component

Equal Component = Alla R och C lika. Ställer A. Vanligen första och andra ordningens sektioner

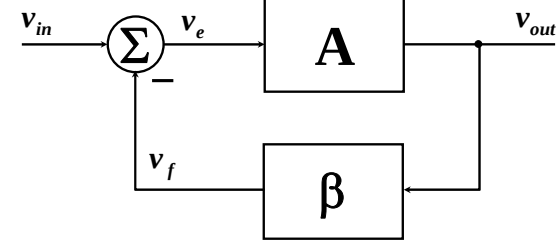
Poler	Steg Unity-gain	Steg Equal-Comp.
2	1	1
3	1	2(1+2)
4	2	2
5	2	3
6	3	3
7	3	4



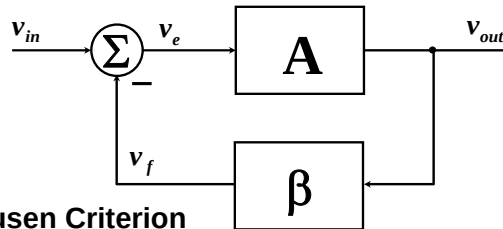
Oscillatorer

Negativ Återkoppling

$$A_{CL} = \frac{A}{1 + \beta A} = [om \ \beta A \gg 1] \approx \frac{1}{\beta}$$

Positiv Återkoppling

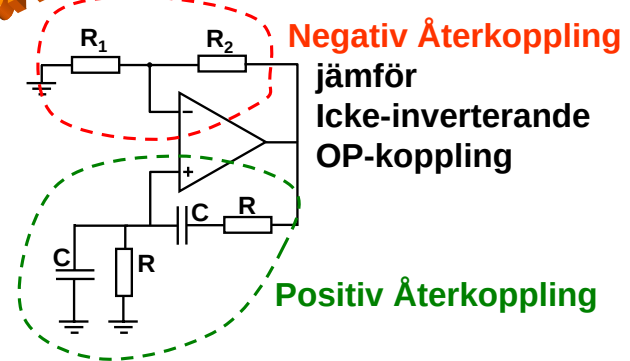
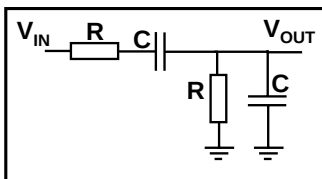
$$A_{CL} = \frac{A}{1 + \beta A} \quad [om \ \beta A = -1] \Rightarrow \text{Självsvängning Oscillation}$$

Positiv ÅterkopplingBarkhausen Criterion

$$\begin{cases} |\beta A| = 1 \\ \varphi = 180^\circ \end{cases}$$

Obs! Giltigt för denna modell!

Kapitel 10.1

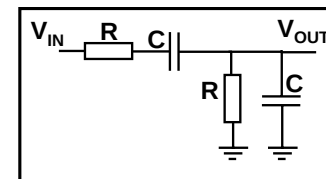
Wien OscillatorWien Filter – $H(s)$ 

$$R \parallel C = \frac{R \cdot \frac{1}{sC}}{R + \frac{1}{sC}} = \frac{R}{1 + sRC}$$

$$H(s) = \frac{R \parallel C}{R + \frac{1}{sC} + R \parallel C}$$

$$H(s) = \frac{\frac{R}{1 + sRC}}{R + \frac{1}{sC} + \frac{R}{1 + sRC}}$$

$$H(s) = \frac{sRC}{(sRC)^2 + s3RC + 1}$$

Wien Filter – $H(s)$ 

$$H(s) = \frac{sRC}{(sRC)^2 + s3RC + 1}$$

$$H(s) = \frac{1}{(RC)^2} \frac{sRC}{s^2 + \frac{s3}{RC} + \frac{1}{(RC)^2}}$$

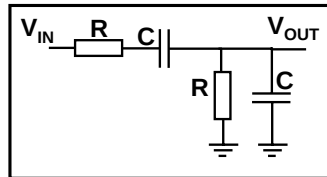
Finn rötterna till

$$s^2 + \frac{s3}{RC} + \frac{1}{(RC)^2}$$

vilket är polerna till $H(s)$

$$p_{1,2} = -\frac{3}{2RC} \pm \sqrt{\left(\frac{3}{2RC}\right)^2 - \frac{1}{(RC)^2}}$$

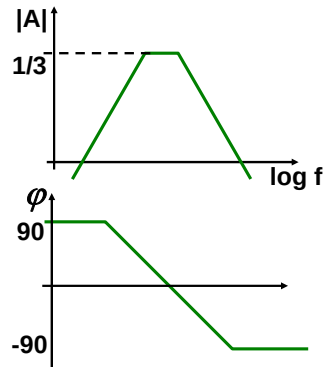
Wien Filter – $H(s)$



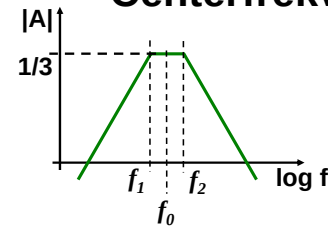
Nollställ i 0

$$H(s) = \frac{1}{(RC)^2} \frac{sRC}{s^2 + \frac{s3}{RC} + \frac{1}{(RC)^2}}$$

2 poler



Centerfrekvens – Wien-filter



$$\text{Centerfrekvens: } \omega_0 = \sqrt{\omega_1 \omega_2}$$

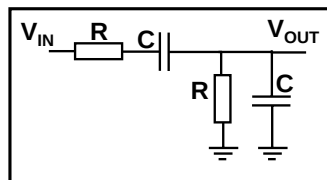
$$p_{1,2} = -\frac{3}{2RC} \pm \sqrt{\left(\frac{3}{2RC}\right)^2 - \frac{1}{(RC)^2}}$$

$$\log \omega_0 = \frac{\log \omega_1 + \log \omega_2}{2}$$

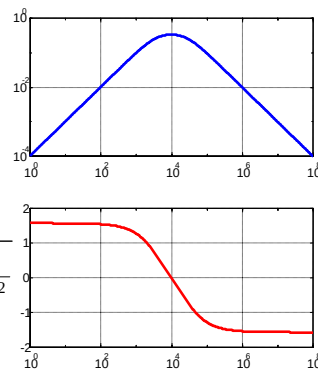
$$\log \omega_0 = \frac{1}{2} \log \omega_1 \omega_2 = \log \sqrt{\omega_1 \omega_2}$$

$$\omega_0 = \frac{1}{RC}$$

Wien Filter – $H(j\omega)$



$$H(j\omega) = \frac{1}{(RC)^2} \frac{j\omega RC}{(j\omega)^2 + \frac{3j\omega}{RC} + \frac{1}{(RC)^2}}$$



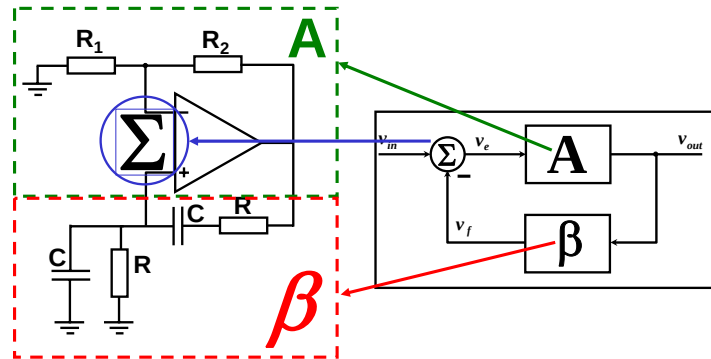
Wien Filter – $H(j\omega)$

Skriv om:

$$H(j\omega) = \frac{1}{(RC)^2} \frac{j\omega RC}{(j\omega)^2 + \frac{3j\omega}{RC} + \frac{1}{(RC)^2}}$$

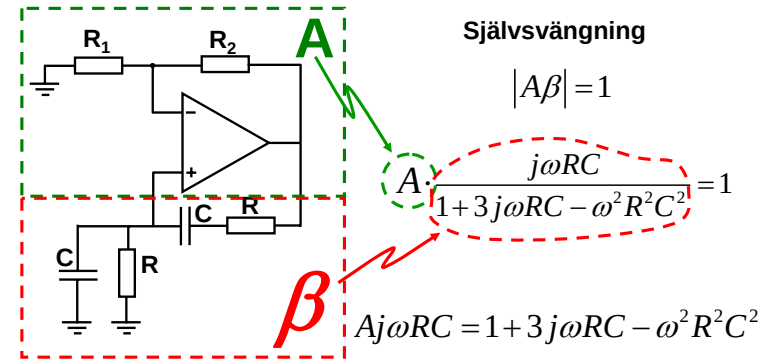
$$H(j\omega) = \frac{j\omega RC}{1 + 3j\omega RC - (\omega RC)^2}$$

Wien Oscillator



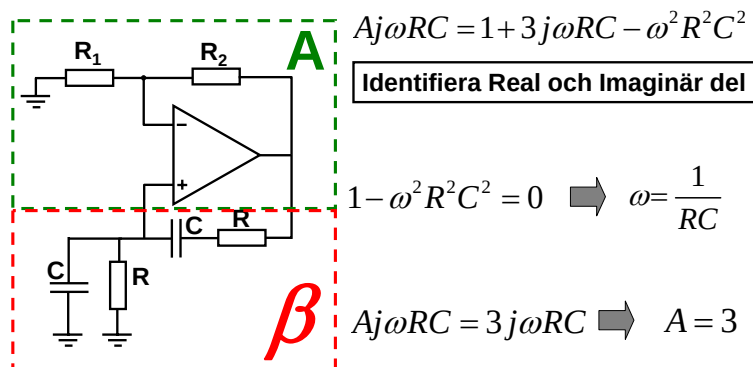
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Wien Oscillator



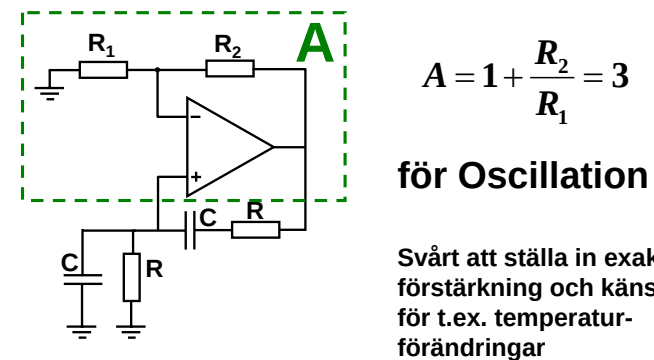
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Wien Oscillator



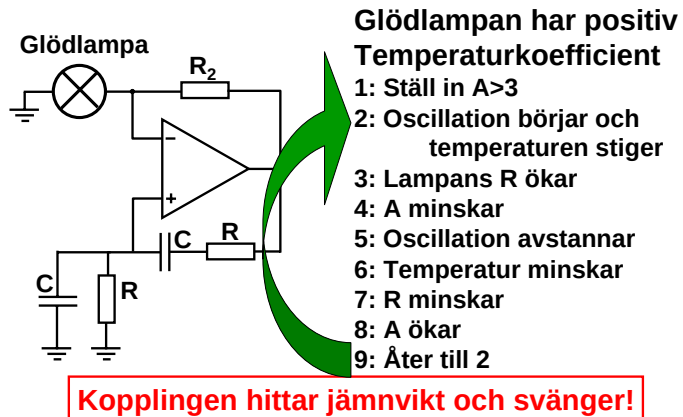
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Wien Oscillator



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Wien Oscillator med Stabilisering



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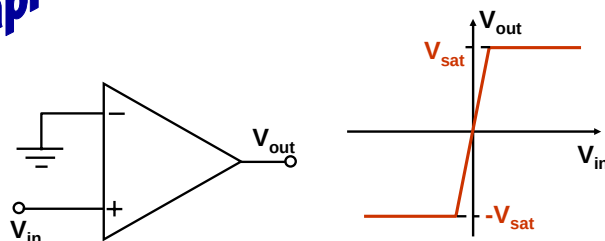
Komparatorn och Relaxationsoscillatorn



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Kap. 9

Icke-inverterande Komparator

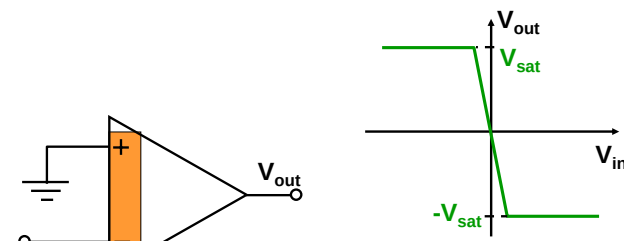


Ingen Återkoppling!
Utgången antar ett av sina extremlägen, V_{sat} eller $-V_{sat}$

Idealt : $\frac{dV_{out}}{dt} = \infty$

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Inverterande Komparator

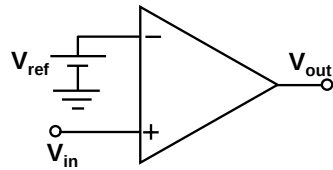


Bytt plats på Ingångarna.

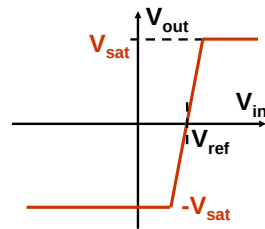
Idealt : $\frac{dV_{out}}{dt} = \infty$

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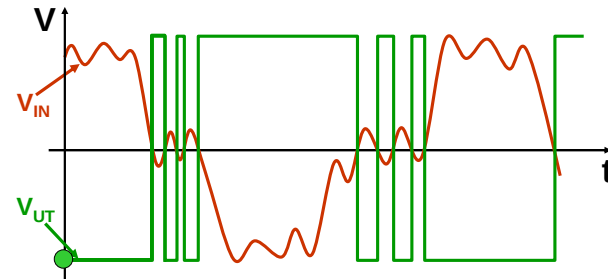
Icke-inverterande Komparator



Jämför med en referensspänning, V_{ref} , istället för GND.

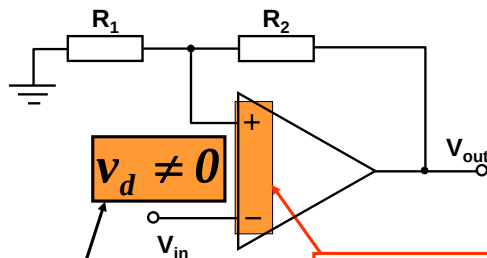


Störningar och Komparator



Distorderad signal kan orsaka multipla oönskade övergångar vilket ger ökad effektförbrukning och störningar.

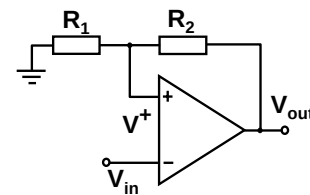
Komparator med Hysteres



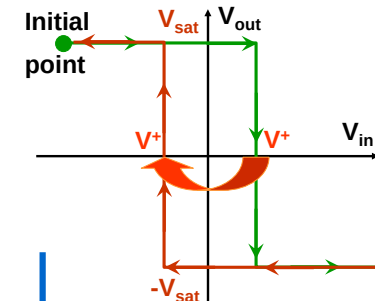
Positiv Återkoppling

Bytt plats på ingångarna gentemot Icke-inverterande OP:koppling

Komparator med Hysteres

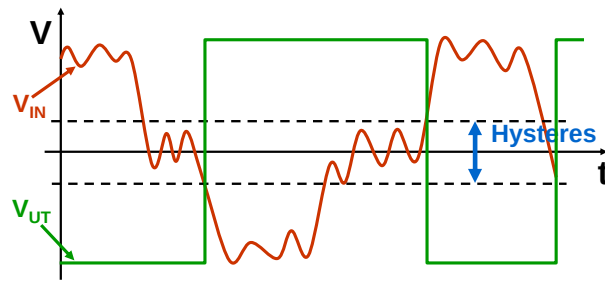


1. Initialt stor negativ V_{in}
2. Inverterande $\Rightarrow V_{out} = V_{sat}$
3. $V^+ = \frac{R_1}{R_1 + R_2} V_{sat} (V_{REF})$
4. När V_{in} ökar slår komparatorn om och samtidigt ändras V^+



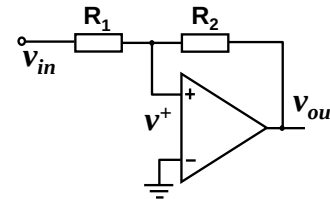
5. $V^+ = \frac{R_1}{R_1 + R_2} (-V_{sat}) (V_{REF})$
6. När V_{in} minskar slår komparatorn om vid ny V^+

Störningar och Komparator



Hysteres minskar antalet övergångar!

Icke-Inverterande Komparator med Hysteres



v^+ är en linjärkombination
av v_{out} och v_{in}

Komparatorn slår om när
 v^+ passerar 0

Bidrag till v^+

Bidrag från v_{in} - Nollställ v_{out}

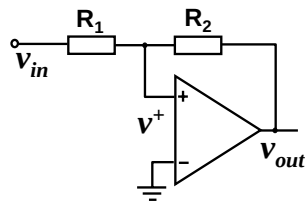
$$v^{+in} = \frac{R_2}{R_1 + R_2} v_{in}$$

Bidrag från v_{out} - Nollställ v_{in}

$$v^{+out} = \frac{R_1}{R_1 + R_2} v_{out}$$

V_{sat}

Icke-inverterande Schmitt Trigger



$$v^+ = \frac{R_2}{R_1 + R_2} v_{in} + \frac{R_1}{R_1 + R_2} v_{out}$$

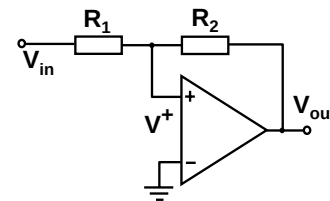
Om v_{in} och v_{out} negativ

$$v^+ = \frac{R_2}{R_1 + R_2} v_{in} - \frac{R_1}{R_1 + R_2} V_{sat}$$

Omslag vid $v^+ = 0$

$$v_{in} > \frac{R_1}{R_2} V_{sat}$$

Komparator med Hysteres

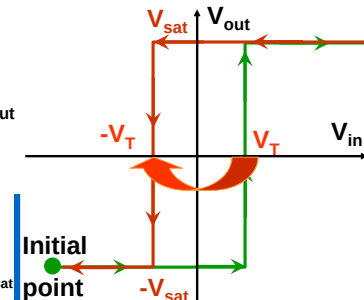


1. Initialt stor negativ V_{in}

2. Icke- Inverterande $\Rightarrow V_{out} = -V_{sat}$

$$3. V_T = \frac{R_1}{R_2} V_{sat}$$

4. När V_{in} ökar slår komparatorn om och samtidigt ändras V_T

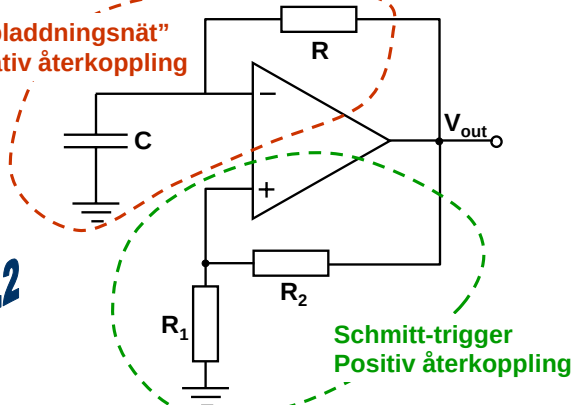


$$5. V_T = -\frac{R_1}{R_2} V_{sat}$$

6. När V_{in} minskar slår komparatorn om vid ny V_T

Multivibrator - Relaxationsoscillator

"Uppladdningsnät"
Negativ återkoppling

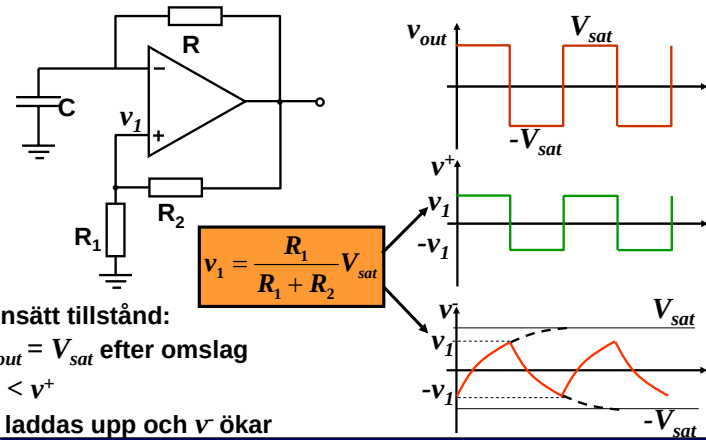


kap. 10.2

Schmitt-trigger
Positiv återkoppling

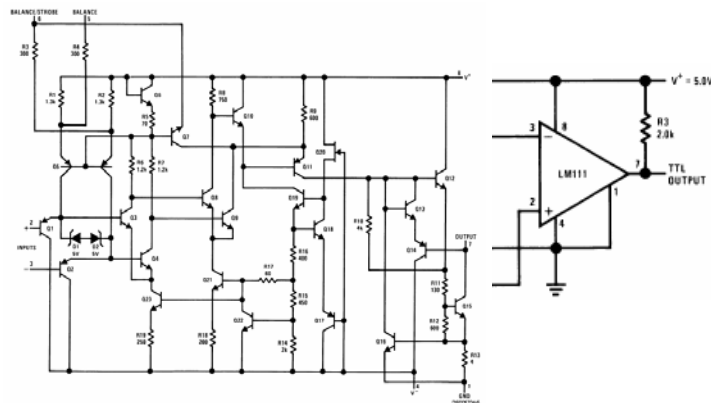
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Multivibrator - Relaxationsoscillator



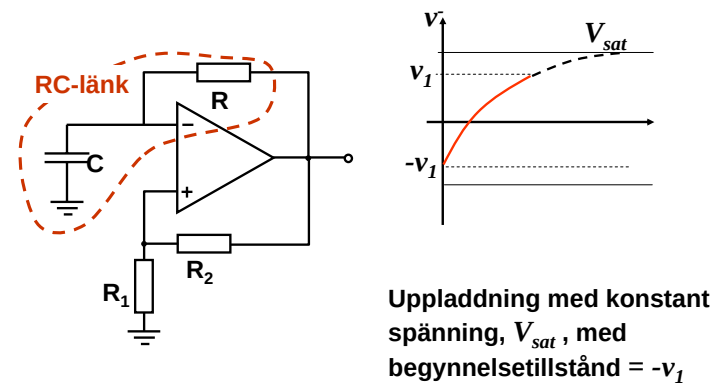
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Open Collector

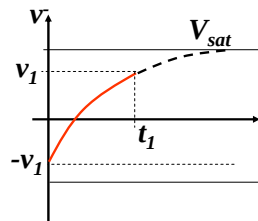


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Multivibrator - Relaxationsoscillator



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Multivibrator - Relaxationsoscillator

$$v = Ae^{-t/RC} + V_{sat}(1 - e^{-t/RC})$$

$$v(t=0) = -v_1 \Rightarrow A = -v_1$$

$$v(t=t_1) = v_1$$

$$v_1 = V_{sat} - (v_1 + V_{sat})e^{-t_1/RC}$$

Uppladdning med konstant
spänning, V_{sat} , med
begynnelsestillstånd = $-v_1$
(Jämför Stegsvär!)

Multivibrator - Relaxationsoscillator

$$v_1 = V_{sat} - (v_1 + V_{sat})e^{-t_1/RC}$$

$$t_1 = \tau \ln \left(\frac{V_{sat} + v_1}{V_{sat} - v_1} \right)$$

$$v_1 = \frac{R_1}{R_1 + R_2} V_{sat}, \quad T = 2t_1 \Rightarrow T = 2\tau \ln \left(1 + \frac{2R_1}{R_2} \right)$$

